

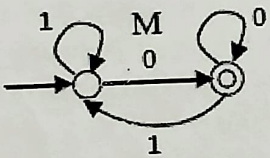
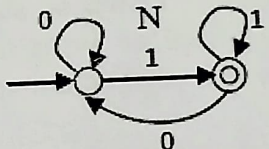
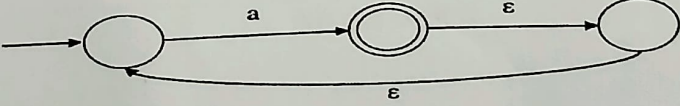
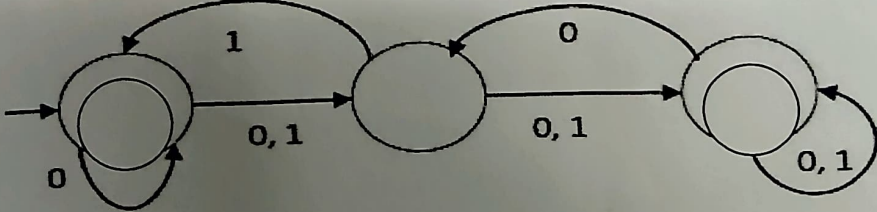
Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL**Department of Computer Science & Engineering (CSE)****Mid-Term Examination September - 2025**Subject Name: **Formal Language & Automata Theory**
Subject Code: **CSE/AI/CPS/CS/DS-3004**Time Allotted: **90 min**
MM:20

NOTE: 1. All questions are compulsory.

2. Mark answer number clearly.

1.	Construct DFA for the following languages with minimum number of states (a) Consider the set of strings on $\{0,1\}$ in which, every substring of 3 symbols has at most two zeros. For examples, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. (b) All the strings of $\{a, b\}$ in which last two symbols are not distinct.	3+2
2.	(a) Consider the regular language $L = (aaa + aaaaa)^*$. Construct minimum state DFA that recognizes L (b) Consider the DFAs M and N given below   Find the minimum state DFA that accepts $(L(M) \cap L(N))$.	3+2
3.	(a) Construct a DFA equivalent to the following regular expressions $((aa + bb + \epsilon)(ab + ba)^* + a)^* + a$ (b) Design a Moore Machine to invert the Binary string?	3+2
4.	(a) What is the complement of the language accepted by the NFA shown below? Assume $\Sigma = \{a\}$ and ϵ is the empty string  (b) Consider the DFA M shown below find regular expression accepted by this automaton 	3+2